Ex: 9

Date:

Implementation of Subnetting in CISCO PACKET TRACKER Simulator

a) Design Multiple Subnets with a suitable number of host

AIM:

To design Multiple subnets with a suitable number of hosts using sub netting techniques in cisco packets Trainer.

STEPS:

1) Open cisco packet Tracer and create a new project.
2) Design the network topology by planning the required network device.
3) Select a suitable network address for the experiment.
4) calculate the number of Subnets required.
5) Determine the no. of host needed in each subnet.
6) calculate the Subnet mask using subnetting techniques
7) Divide the network into multiple subnets.
8) Assign a separate network address to each Subnet.
9) verify the Subnet calculations and host ranges
10) Save the network topology.

RESULT:

Thus, multiple subnet were successfully designed according to the required number of hosts.

IP configuration
Start

IP address 192.168.1.11
Subnet mask 255.255.255.0
Default Gateway 192.168.1.1

IPV6 configuration
Link local address FE80::1:2DD:BAFF:FE9:D809

b) Connect the subnets through a router.

AIM:- To assign static IP address to all devices in different subnets and establish communication between the subnet through a routes.

STEPS:-

1) Connect a router in the 3 switches using appropriate cables.
2) Configure the router interfaces.
3) Assign an IP address & subnet mask to each other router
4) Configure the Switches.
5) Assign static IP address to all computer.
6) Configure the Subnet mask for each device.
7) Assign appropriate default gateway to each computer
8) Verify the IP configuration.
9) ~~Verify the IP configuration~~ Save the router configuration.
10) Verify the all devices are properly connected.

RESULT:-

Thus, the static IP addresses were assigned Successfully and communicated between the subnets was established through the router.

PC0

Physical Config Desktop Custom Interface

Command Prompt

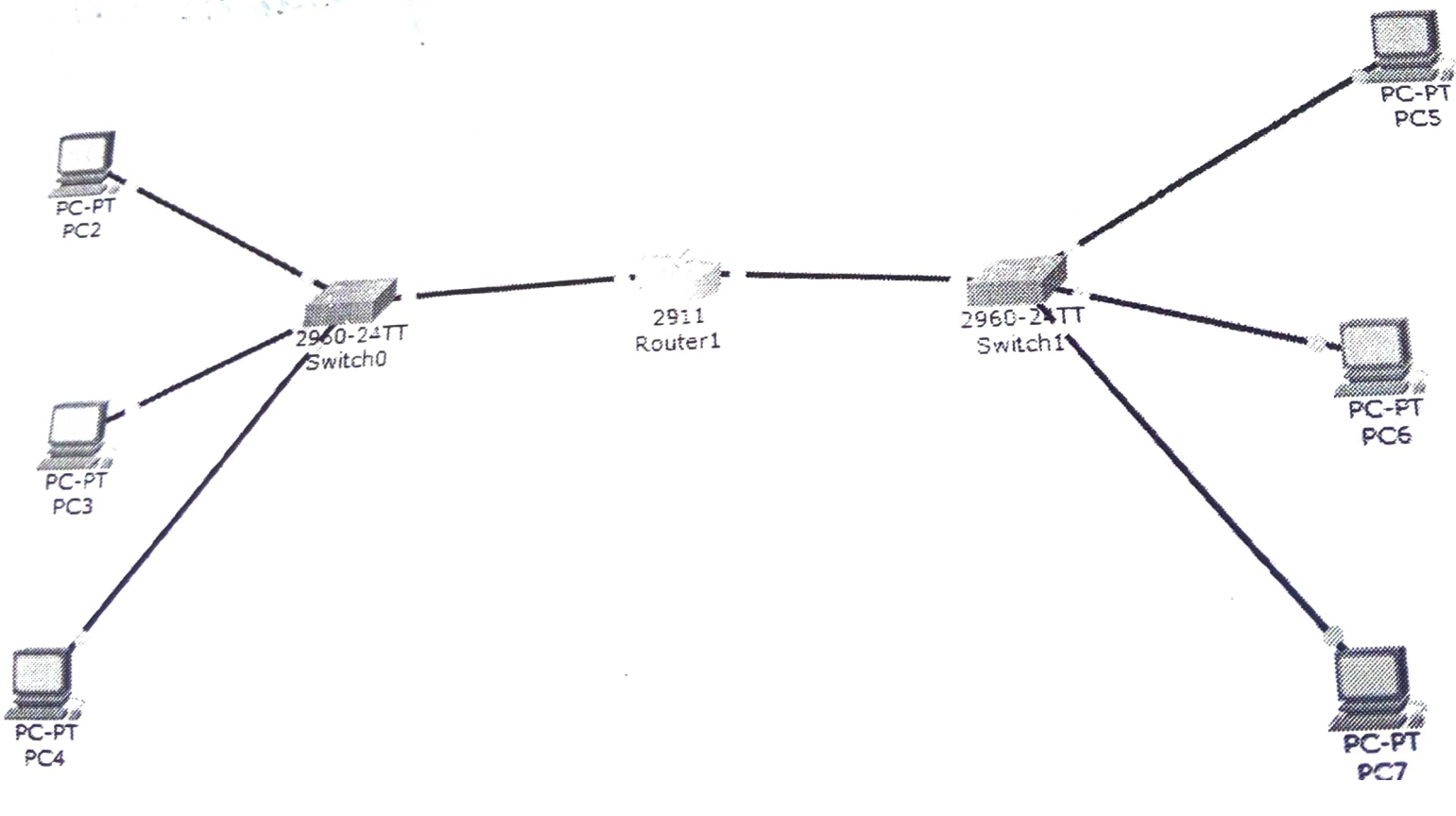
```
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

PC>
```



Simulate packet Transmission accross the subnets and observe the result.

(a) when Subnets are Connected through a Router.

AIM:

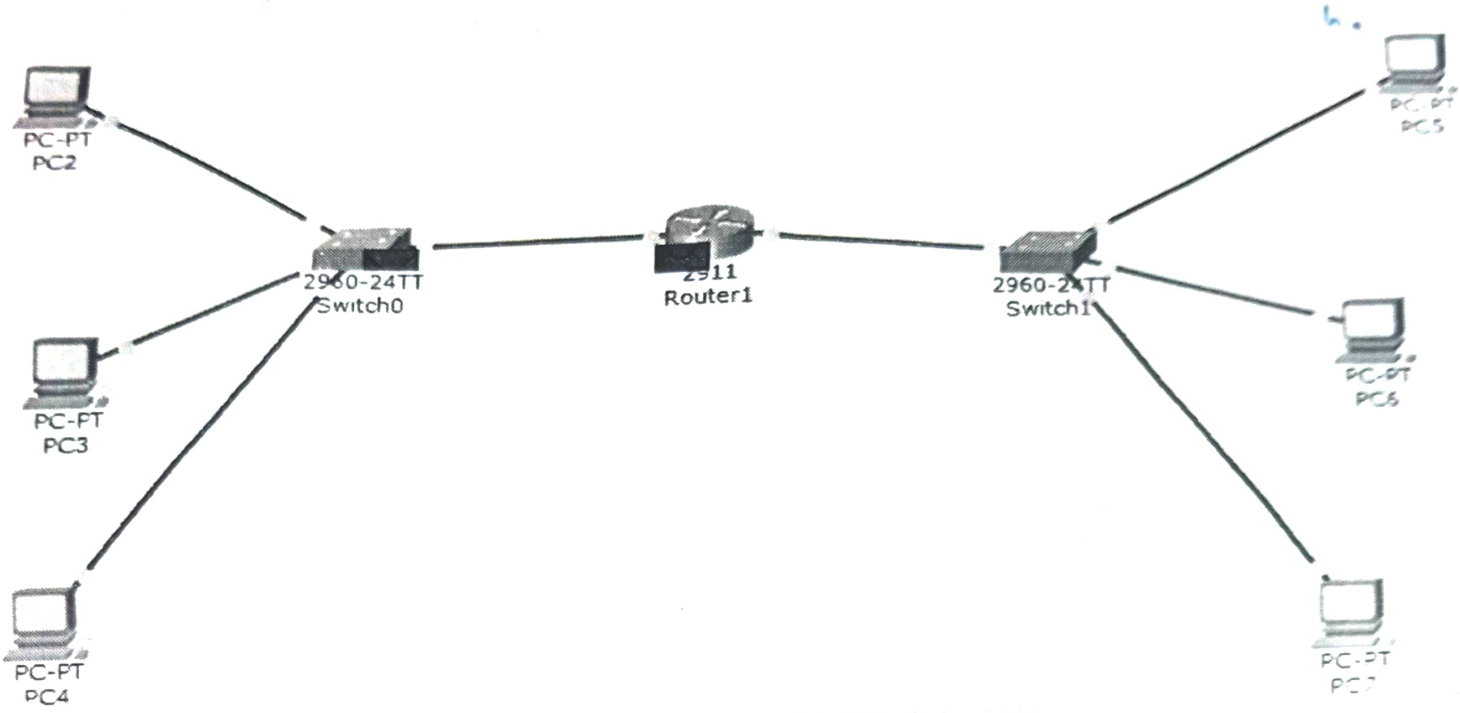
To analyze Packet Transmission between different Subnets when a router when a router is used for communication when a router is used for communication.

STEPS:

1) open simulator mode in cisco packet Tracer.
2) select a source device from one Subnet.
3) Select a destination device from another subnet.
4) Generate a packet using simple PDU tool.
5) observe movement of packet through the network.
6) Analyze how the router receives the packet.
7) Observe forwarding process performed by router.
8) verify the packet info between different Subnets.
9) use the ping command to test Connectivity.
10) observe the successful delivery of packets.
11) Records the simulation Results.
12) Analyze the routing process.

RESULT:

Thus, packets were transmitted successfully between different subnets through the router.


PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
2960-24TT
Switch0
2911
Router1
2960-24TT
Switch1
PC-PT
PC5
PC-PT
PC6